Adam McMillan

Software & Machine Learning Engineer

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Profile

Software and machine learning developer with experience delivering production tools, modernising legacy applications and conducting original AI research. Combines Python and React/Django engineering with statistical ML, optimisation and research communication; independently built ML automation that reduced estimated data-onboarding time by over 80%.

Technical Skills

Languages and data: Python, JavaScript, SQL, C++, NumPy, pandas
Software engineering: React, Django, REST APIs, relational databases, Git, CI/CD
AI/ML and research: scikit-learn, PyTorch, embeddings, MLPs, CVXPY, statistical learning, model evaluation
Infrastructure: Google Cloud, Docker, Kubernetes, Linux/Unix

Professional Experience

Software Developer (Intern; later Casual) — Exergenics *Jul–Oct 2025; Dec 2025–Mar 2026*

- Reduced estimated HVAC data onboarding time by over 80% (roughly 30 to under 5 minutes per dataset) by deploying an ML-assisted pipeline that mapped messy customer CSV headers to standardised fields for engineer review.
- Designed hierarchical classifiers using OpenAI embeddings and MLPs, selected models through validation-set grid search and held-out testing, and combined outputs with equipment metadata and rule-based processing.
- Independently migrated building and plant configuration from undocumented PHP into React/Django, designing APIs and database integrations for production tools used regularly by engineers.
- Owned feature delivery from requirements and implementation to Git-based CI/CD, Google Cloud deployment and user-feedback iterations in a team of fewer than 10.

Research Assistant (Software Engineering) — Griffith University *Jul–Nov 2026*

- Modernised a roughly 20-year-old digital music research archive for public web release, replacing obsolete QuickTime/Flash functionality and rebuilding an interactive score viewer with OpenSeadragon.
- Implemented responsive HTML/CSS/JavaScript interfaces, restored archival media and resolved legacy compatibility issues across browsers.

Tutor, Statistical Machine Learning (COMP90051) — University of Melbourne *Semester 2, 2026*

- Taught approximately 25 postgraduate students across 12 weekly tutorials, explaining mathematical ML concepts and guiding Python/Jupyter implementations; assessed assignments, examinations and research projects.

Selected Research

Master's research: Latent-Group Methods for Exploratory Fair Classifier Learning *2026 | 91 (H1)*

- Developed Spectral-Z and Dual-Z for exploratory fairness auditing without predefined sensitive attributes, including constrained convex optimisation of classifier parameters and latent group assignments.
- Evaluated synthetic data and four real-world datasets; in within-sample COMPAS experiments, measured a selection-rate gap of 0.043 for Dual-Z versus 0.676 for a logistic baseline, using inferred audit groups rather than verified demographics.

Supervisor: Jean Honorio; manuscript in preparation.

Education

Master of Information Technology (Artificial Intelligence) — University of Melbourne *Completed Nov 2026*
H1 average (approximately 87); Dean's Honours List 2025; Research Project 91; Statistical Machine Learning 91; Programming & Software Development 96.

Bachelor of Psychology (Honours) — Deakin University *2023*
Class I Honours; honours research supervised by Dr Pamela Barhoun in the Cognitive Neuroscience Unit.

Bachelor of Music (Honours) — Griffith University *2014*
Class I Honours; University Medal.